

Business Problem Statement

Customer Spending & Loyalty Analysis

Background

Retail teams often have customer shopping data but do not have a short, reproducible analysis that explains where spending is concentrated or which customer behaviors are worth investigating. This project uses a 3,900-record shopping dataset to create a practical analysis in Python and MySQL.

Problem to solve

Identify useful descriptive patterns in category spend, customer groups, shipping options, reviews, locations, and prior purchases. The output should help a student analyst explain what the data can support, what it cannot support, and which questions should be tested next.

Stakeholders

Stakeholder	Question they may ask	Project output
Merchandising	Which categories and products need attention?	Category and product summaries
Customer team	Which groups are candidates for a retention test?	Loyalty and subscription comparisons
Operations	Are shipping options or locations worth reviewing?	Shipping and location tables
Analyst / recruiter	How was the work done?	Notebook, SQL, report, and presentation

Success criteria

1. Every headline number can be traced back to the included CSV.
2. The notebook can be rerun locally.
3. SQL questions use clear, commented logic.
4. Findings are presented with a limitation or validation step where needed.
5. Charts use appropriate chart types and communicate a clear takeaway.

Scope and boundaries

Included: demographic fields, product/category attributes, purchase amount, reviews, subscription/discount fields, shipping type, payment method, location, prior purchases, and purchase-frequency labels.

Not available: timestamps, margin, campaign cost, inventory, store footprint, or repeated transactions. Therefore, the project does not claim trend, profitability, or causal promotion results.

Planned analysis questions

#	Question
1	What are the core portfolio indicators?
2	Which categories and locations lead recorded spend?
3	How do age, loyalty, subscription, and discount groups compare?
4	Do shipping choices show meaningful differences in average purchase?

#	Question
5	Which products have a combination of demand and positive ratings?
6	What data should be added for stronger future analysis?